

Creation Date 15-Apr-2009

Revision Date 24-Mar-2024

Revision Number 2

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                      |                                  |
|----------------------|----------------------------------|
| Product Description: | <b>Diethyl ether</b>             |
| Cat No. :            | <b>C17683</b>                    |
| Synonyms             | Ethyl ether; Ether               |
| Index No             | 603-022-00-4                     |
| CAS No               | 60-29-7                          |
| EC No                | 200-467-2                        |
| Molecular Formula    | C <sub>4</sub> H <sub>10</sub> O |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Flammable liquids

Category 1 (H224)

### Health hazards

Acute oral toxicity

Category 4 (H302)

Specific target organ toxicity - (single exposure)

Category 3 (H336)

### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### **Hazard Statements**

H224 - Extremely flammable liquid and vapor

H302 - Harmful if swallowed

H336 - May cause drowsiness or dizziness

EUH019 - May form explosive peroxides

EUH066 - Repeated exposure may cause skin dryness or cracking

### **Precautionary Statements**

P240 - Ground and bond container and receiving equipment

P243 - Take action to prevent static discharges

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell

P403 + P235 - Store in a well-ventilated place. Keep cool

## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

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## 3.1. Substances

| Component   | CAS No  | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                      |
|-------------|---------|-------------------|----------|----------------------------------------------------------------------------------------|
| Ethyl ether | 60-29-7 | EEC No. 200-467-2 | >95      | Flam. Liq. 1 (H224)<br>Acute Tox. 4 (H302)<br>STOT SE 3 (H336)<br>(EUH019)<br>(EUH066) |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                       |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                                                                                                                                                                                                             |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                        |
| <b>Inhalation</b>                         | Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                      |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                                                 |
|---------------------------|-------------------------------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. Symptoms may be delayed. |
|---------------------------|-------------------------------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

### 5.2. Special hazards arising from the substance or mixture

Extremely flammable. Risk of ignition. Vapors may travel to source of ignition and flash back. Vapors may form explosive mixtures with air. Containers may explode when heated. May form explosive peroxides. Vapors may form explosive mixtures with air.

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## Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), peroxides.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Avoid contact with skin, eyes or clothing.

### 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### 6.3. Methods and material for containment and cleaning up

Remove all sources of ignition. Soak up with inert absorbent material. Take precautionary measures against static discharges. Keep in suitable, closed containers for disposal. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Handle under an inert atmosphere. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Do not breathe mist/vapors/spray. Keep away from open flames, hot surfaces and sources of ignition. If peroxide formation is suspected, do not open or move container. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Flammables area. Store under an inert atmosphere. Keep away from open flames, hot surfaces and sources of ignition. May form explosive peroxides. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from heat, sparks and flame. Keep container tightly closed in a dry and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

**Switzerland - Storage of hazardous substances**

Storage class - SC 3  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>

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<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component   | European Union                                                                                                       | The United Kingdom                                                                                                 | France                                                                                                                                                                                                                  | Belgium                                                                                                                        | Spain                                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethyl ether | TWA: 100 ppm (8h)<br>TWA: 308 mg/m <sup>3</sup> (8h)<br>STEL: 200 ppm (15min)<br>STEL: 616 mg/m <sup>3</sup> (15min) | STEL: 200 ppm 15 min<br>STEL: 620 mg/m <sup>3</sup> 15 min<br>TWA: 100 ppm 8 hr<br>TWA: 310 mg/m <sup>3</sup> 8 hr | TWA / VME: 100 ppm (8 heures). restrictive limit<br>TWA / VME: 308 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 200 ppm. restrictive limit<br>STEL / VLCT: 616 mg/m <sup>3</sup> . restrictive limit | TWA: 100 ppm 8 uren<br>TWA: 308 mg/m <sup>3</sup> 8 uren<br>STEL: 200 ppm 15 minuten<br>STEL: 616 mg/m <sup>3</sup> 15 minuten | STEL / VLA-EC: 200 ppm (15 minutos).<br>STEL / VLA-EC: 616 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 100 ppm (8 horas)<br>TWA / VLA-ED: 308 mg/m <sup>3</sup> (8 horas) |

| Component   | Italy                                                                                                                                                                                                  | Germany                                                                                                                                                                                                                                                           | Portugal                                                                                                                         | The Netherlands                                                             | Finland                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethyl ether | TWA: 100 ppm 8 ore.<br>Time Weighted Average<br>TWA: 308 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>STEL: 200 ppm 15 minuti. Short-term<br>STEL: 616 mg/m <sup>3</sup> 15 minuti. Short-term | TWA: 400 ppm (8 Stunden). AGW - exposure factor 1<br>TWA: 1200 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 400 ppm (8 Stunden). MAK<br>TWA: 1200 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 400 ppm<br>Höhepunkt: 1200 mg/m <sup>3</sup> | STEL: 200 ppm 15 minutos<br>STEL: 616 mg/m <sup>3</sup> 15 minutos<br>TWA: 100 ppm 8 horas<br>TWA: 308 mg/m <sup>3</sup> 8 horas | STEL: 616 mg/m <sup>3</sup> 15 minuten<br>TWA: 308 mg/m <sup>3</sup> 8 uren | TWA: 100 ppm 8 tunteina<br>TWA: 310 mg/m <sup>3</sup> 8 tunteina<br>STEL: 200 ppm 15 minuutteina<br>STEL: 620 mg/m <sup>3</sup> 15 minuutteina |

| Component   | Austria                                                                                                                                              | Denmark                                                                                                                            | Switzerland                                                                                                                            | Poland                                                                            | Norway                                                                                                                                                                 |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethyl ether | MAK-KZGW: 200 ppm 15 Minuten<br>MAK-KZGW: 600 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 100 ppm 8 Stunden<br>MAK-TMW: 300 mg/m <sup>3</sup> 8 Stunden | TWA: 100 ppm 8 timer<br>TWA: 309 mg/m <sup>3</sup> 8 timer<br>STEL: 616 mg/m <sup>3</sup> 15 minutter<br>STEL: 200 ppm 15 minutter | STEL: 400 ppm 15 Minuten<br>STEL: 1200 mg/m <sup>3</sup> 15 Minuten<br>TWA: 400 ppm 8 Stunden<br>TWA: 1200 mg/m <sup>3</sup> 8 Stunden | STEL: 600 mg/m <sup>3</sup> 15 minutach<br>TWA: 300 mg/m <sup>3</sup> 8 godzinach | TWA: 100 ppm 8 timer<br>TWA: 300 mg/m <sup>3</sup> 8 timer<br>STEL: 150 ppm 15 minutter. value calculated<br>STEL: 375 mg/m <sup>3</sup> 15 minutter. value calculated |

| Component   | Bulgaria                                                                                     | Croatia                                                                                                                                                    | Ireland                                                                                                              | Cyprus                                                                                     | Czech Republic                                                           |
|-------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Ethyl ether | TWA: 100 ppm<br>TWA: 308 mg/m <sup>3</sup><br>STEL : 200 ppm<br>STEL : 616 mg/m <sup>3</sup> | TWA-GVI: 100 ppm 8 satima.<br>TWA-GVI: 308 mg/m <sup>3</sup> 8 satima.<br>STEL-KGVI: 200 ppm 15 minutama.<br>STEL-KGVI: 616 mg/m <sup>3</sup> 15 minutama. | TWA: 100 ppm 8 hr.<br>TWA: 308 mg/m <sup>3</sup> 8 hr.<br>STEL: 200 ppm 15 min<br>STEL: 616 mg/m <sup>3</sup> 15 min | STEL: 200 ppm<br>STEL: 616 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 308 mg/m <sup>3</sup> | TWA: 300 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 600 mg/m <sup>3</sup> |

| Component | Estonia | Gibraltar | Greece | Hungary | Iceland |
|-----------|---------|-----------|--------|---------|---------|
|-----------|---------|-----------|--------|---------|---------|

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|             |                                                                                                                                                |                                                                                                                    |                                                                                              |                                                                                                                                    |                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Ethyl ether | TWA: 100 ppm 8 tündides.<br>TWA: 308 mg/m <sup>3</sup> 8 tündides.<br>STEL: 200 ppm 15 minutites.<br>STEL: 616 mg/m <sup>3</sup> 15 minutites. | TWA: 100 ppm 8 hr<br>TWA: 308 mg/m <sup>3</sup> 8 hr<br>STEL: 200 ppm 15 min<br>STEL: 616 mg/m <sup>3</sup> 15 min | STEL: 500 ppm<br>STEL: 1500 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 1200 mg/m <sup>3</sup> | STEL: 616 mg/m <sup>3</sup> 15 percekben. CK<br>TWA: 308 mg/m <sup>3</sup> 8 órában. AK<br>lehetséges borön keresztüli felszívódás | STEL: 200 ppm<br>STEL: 616 mg/m <sup>3</sup><br>TWA: 100 ppm 8 klukkustundum.<br>TWA: 308 mg/m <sup>3</sup> 8 klukkustundum. |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|

| Component   | Latvia                                                                                     | Lithuania                                                                                            | Luxembourg                                                                                                                           | Malta                                                                                                          | Romania                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Ethyl ether | STEL: 200 ppm<br>STEL: 616 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 308 mg/m <sup>3</sup> | TWA: 308 mg/m <sup>3</sup> IPRD<br>TWA: 100 ppm IPRD<br>STEL: 616 mg/m <sup>3</sup><br>STEL: 200 ppm | TWA: 100 ppm 8 Stunden<br>TWA: 308 mg/m <sup>3</sup> 8 Stunden<br>STEL: 200 ppm 15 Minuten<br>STEL: 616 mg/m <sup>3</sup> 15 Minuten | TWA: 100 ppm<br>TWA: 308 mg/m <sup>3</sup><br>STEL: 200 ppm 15 minuti<br>STEL: 616 mg/m <sup>3</sup> 15 minuti | TWA: 100 ppm 8 ore<br>TWA: 308 mg/m <sup>3</sup> 8 ore<br>STEL: 200 ppm 15 minute<br>STEL: 616 mg/m <sup>3</sup> 15 minute |

| Component   | Russia                                                        | Slovak Republic                                                              | Slovenia                                                                                                                       | Sweden                                                                                                                                                       | Turkey                                                                                                                       |
|-------------|---------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Ethyl ether | TWA: 300 mg/m <sup>3</sup> 2469<br>MAC: 900 mg/m <sup>3</sup> | Ceiling: 616 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 308 mg/m <sup>3</sup> | TWA: 100 ppm 8 urah<br>TWA: 308 mg/m <sup>3</sup> 8 urah<br>STEL: 200 ppm 15 minutah<br>STEL: 616 mg/m <sup>3</sup> 15 minutah | Binding STEL: 200 ppm 15 minuter<br>Binding STEL: 616 mg/m <sup>3</sup> 15 minuter<br>TLV: 100 ppm 8 timmar. NGV<br>TLV: 308 mg/m <sup>3</sup> 8 timmar. NGV | TWA: 100 ppm 8 saat<br>TWA: 308 mg/m <sup>3</sup> 8 saat<br>STEL: 200 ppm 15 dakika<br>STEL: 616 mg/m <sup>3</sup> 15 dakika |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                    | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Ethyl ether<br>60-29-7 (>95) |                              |                                 |                                | DNEL = 44mg/kg bw/day             |

| Component                    | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Ethyl ether<br>60-29-7 (>95) |                                  | DNEL = 616mg/m <sup>3</sup>         |                                    | DNEL = 308mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

See values below.

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| Component                      | Fresh water  | Fresh water sediment            | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)          |
|--------------------------------|--------------|---------------------------------|--------------------|------------------------------------|-----------------------------|
| Ethyl ether<br>60-29-7 ( >95 ) | PNEC = 2mg/L | PNEC = 9.14mg/kg<br>sediment dw | PNEC = 1.65mg/L    | PNEC = 4.2mg/L                     | PNEC = 0.66mg/kg<br>soil dw |

| Component                      | Marine water   | Marine water sediment               | Marine water Intermittent | Food chain | Air |
|--------------------------------|----------------|-------------------------------------|---------------------------|------------|-----|
| Ethyl ether<br>60-29-7 ( >95 ) | PNEC = 0.2mg/L | PNEC =<br>0.914mg/kg<br>sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard       | Glove comments                                                                                                                  |
|----------------|-------------------|-----------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Nitrile rubber | < 33 minutes      | 0.28 - 0.35 mm  | EN 374<br>Level 2 | Permeation rate 36 µg/cm <sup>2</sup> /min<br>As tested under EN374-3 Determination of<br>Resistance to Permeation by Chemicals |
| Viton (R)      | < 19 minutes      | 0.3 mm          |                   |                                                                                                                                 |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

ALFAAC17683

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|                                         |                                                 |                                          |
|-----------------------------------------|-------------------------------------------------|------------------------------------------|
| Physical State                          | Liquid                                          |                                          |
| Appearance                              | Colorless                                       |                                          |
| Odor                                    | aromatic                                        |                                          |
| Odor Threshold                          | No data available                               |                                          |
| Melting Point/Range                     | -116 °C / -176.8 °F                             |                                          |
| Softening Point                         | No data available                               |                                          |
| Boiling Point/Range                     | 34.6 °C / 94.3 °F                               |                                          |
| Flammability (liquid)                   | Extremely flammable                             | On basis of test data                    |
| Flammability (solid,gas)                | Not applicable                                  | Liquid                                   |
| Explosion Limits                        | <b>Lower</b> 1.7 vol %<br><b>Upper</b> 48 vol % |                                          |
| Flash Point                             | -45 °C / -49 °F                                 | <b>Method</b> - No information available |
| Autoignition Temperature                | 160 °C / 320 °F                                 |                                          |
| Decomposition Temperature               | No data available                               |                                          |
| pH                                      | No information available                        |                                          |
| Viscosity                               | 0.2448 cP at 20 °C                              |                                          |
| Water Solubility                        | 69 g/L (20°C)                                   |                                          |
| Solubility in other solvents            | No information available                        |                                          |
| Partition Coefficient (n-octanol/water) |                                                 |                                          |
| Component                               | <b>log Pow</b>                                  |                                          |
| Ethyl ether                             | 0.82                                            |                                          |
| Vapor Pressure                          | 587 mbar @ 20 °C                                |                                          |
| Density / Specific Gravity              | 0.714                                           |                                          |
| Bulk Density                            | Not applicable                                  | Liquid                                   |
| Vapor Density                           | 2.55                                            | (Air = 1.0)                              |
| Particle characteristics                | Not applicable (liquid)                         |                                          |

## 9.2. Other information

|                      |                                             |
|----------------------|---------------------------------------------|
| Molecular Formula    | C4 H10 O                                    |
| Molecular Weight     | 74.12                                       |
| Explosive Properties | Vapors may form explosive mixtures with air |
| Evaporation Rate     | 37.5 - (Butyl Acetate = 1.0)                |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

May form explosive peroxides. Air sensitive. Light sensitive. Hygroscopic.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | May form explosive peroxides.            |

### 10.4. Conditions to avoid

Incompatible products. Heat, flames and sparks. Exposure to air. Exposure to light.  
Exposure to moisture. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Strong oxidizing agents. Strong acids.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). peroxides.



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## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component   | LD50 Oral        | LD50 Dermal       | LC50 Inhalation       |
|-------------|------------------|-------------------|-----------------------|
| Ethyl ether | 1215 mg/kg (Rat) | 20 mL/kg (Rabbit) | 32000 ppm ( Rat ) 4 h |

##### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

##### (c) serious eye damage/irritation;

Based on available data, the classification criteria are not met

##### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

##### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

Mutagenic effects have occurred in experimental animals

##### (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

##### (g) reproductive toxicity;

Based on available data, the classification criteria are not met

##### (h) STOT-single exposure;

Category 3

Results / Target organs

Central nervous system (CNS).

##### (i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

##### (j) aspiration hazard;

Based on available data, the classification criteria are not met

#### Other Adverse Effects

See actual entry in RTECS for complete information

#### Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

### 11.2. Information on other hazards

#### Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

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## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Do not empty into drains.

| Component   | Freshwater Fish                                                                                                   | Water Flea          | Freshwater Algae |
|-------------|-------------------------------------------------------------------------------------------------------------------|---------------------|------------------|
| Ethyl ether | LC50: > 10000 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 2560 mg/L, 96h flow-through (Pimephales promelas) | EC50 = 165 mg/L/24h |                  |

| Component   | Microtox                | M-Factor |
|-------------|-------------------------|----------|
| Ethyl ether | EC50 = 5600 mg/L 15 min |          |

### 12.2. Persistence and degradability

#### Persistence

Persistence is unlikely, based on information available.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Ethyl ether | 0.82    | No data available             |

### 12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

### 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant

#### Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Waste codes should be assigned by the user based on the application for which the product

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was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

## Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

**14.1. UN number** UN1155  
**14.2. UN proper shipping name** Diethyl ether  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** I

### ADR

**14.1. UN number** UN1155  
**14.2. UN proper shipping name** Diethyl ether  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** I

### IATA

**14.1. UN number** UN1155  
**14.2. UN proper shipping name** Diethyl ether  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** I

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component   | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-------------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Ethyl ether | 60-29-7 | 200-467-2 | -      | -   | X     | X    | KE-27690 | X    | X    |

| Component   | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-------------|---------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Ethyl ether | 60-29-7 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Not applicable

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| Component   | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ethyl ether | 60-29-7 | -                                                                   | -                                                                             | -                                                                                                     |

## Seveso III Directive (2012/18/EC)

| Component   | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Ethyl ether | 60-29-7 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component   | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-------------|---------------------------------------|-------------------------|
| Ethyl ether | WGK1                                  |                         |

| Component   | France - INRS (Tables of occupational diseases)      |
|-------------|------------------------------------------------------|
| Ethyl ether | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                    | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Ethyl ether<br>60-29-7 (>95) |                                                                                                                | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

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## Full text of H-Statements referred to under sections 2 and 3

H224 - Extremely flammable liquid and vapor  
H302 - Harmful if swallowed  
H336 - May cause drowsiness or dizziness  
EUH019 - May form explosive peroxides  
EUH066 - Repeated exposure may cause skin dryness or cracking

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

## Prepared By

Health, Safety and Environmental Department

## Creation Date

15-Apr-2009

## Revision Date

24-Mar-2024

## Revision Summary

New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information

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relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**